Wheels and Brakes (Mod 6/2002) – General Data

1. General

   A. Wheels

      The main landing gear wheels are carried on the axle of each main gear leg, and the nose wheel is carried on the axle mounted in the fork of the nose gear. The wheels are of the split hub type to facilitate removal and installation of tubeless tires.

   B. Wheel Brakes System

      The wheel brakes system consists of hydraulically-operated wheel brake units controlled from the pilot's and co-pilot's rudder pedals. The fluid pressure is metered to the wheel brake units in proportion to brake pedal depression. When the hydraulic system electric motor-driven pump or handpump is operated, fluid is supplied from the main hydraulic system to the control valves and to charge a wheel brakes accumulator (refer to 29-10-00, Hydraulic Power Package (Mod 6/2039) – General Data). Return fluid from the control valves is fed back to the hydraulic system common return line. A check valve is installed upstream of the supply line to the accumulator. The accumulator supplies brake pressure when the pumps are not in operation.

   C. Brake Hydraulic Pressure Indicator

      The brake pressure transducers send a signal to display the amount of hydraulic pressure on the cockpit multi-function display screen.

2. Servicing

   A. Bleed Hydraulic Brake System

      1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.). Check nose wheel steering lever corresponds with position of nose wheel, and wing flaps selector with position of wing flaps. Ensure parking brake is OFF.

      2) Set POWER SOURCE switch to EXTERNAL, and MASTER switch to ON (Pre Mod 6/2314) or set EXTERNAL switch to ON (Post Mod 6/2314). Check electric motor-driven hydraulic pump charges damping accumulator and wheel brakes accumulator, and motor cuts out when reading on pressure indicators is 1575 ± 50 psi.

      3) Depress both left and right brake pedals and engage the parking brake.

      4) Place container to receive spillage, and slacken bleeder plug at top rear end of left brake unit. When fluid is clear of air, retighten bleeder plug.

      5) Repeat Para 2.A.4) for right wheel brake unit.

      6) Release parking brake.

      7) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

      8) Check Hydraulic reservoir fluid level.

3. Adjustment/Test

   A. Function Test Wheel Brakes

      1) Ensure parking brake is OFF.

      2) Remove rear bleeder plug at top rear end of left and right-hand brake unit, and install 0 to 2000 psi pressure gauge at each bleeder port.

3) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.). Discharge brake system accumulator hydraulic pressure by pumping brake pedals. Check nose wheel steering lever corresponds with position of nose wheel, and wing flaps selector with position of wing flaps.

4) Set POWER SOURCE switch to EXTERNAL, and MASTER switch to ON (Pre Mod 6/2314) or set EXTERNAL switch to ON (Post Mod 6/2314). Check electric motor-driven hydraulic pump charges damping accumulator and wheel brakes accumulator, and motor cuts out when reading on pressure indicators is 1575 ± 50 psi.

5) Depress pilot's left brake pedal and check pressure gage reading at left brake unit settles at 1000 ± 50 psi. Release brake pedal. Repeat test using co-pilot's left brake pedal.

6) Depress pilot's right pedal and check pressure gage reading at right brake unit settles at 1000 ± 50 psi. Release brake pedal. Repeat test using co-pilot's right brake pedal.

   Note Adjust pressure if necessary at turnbuckles on push rods.

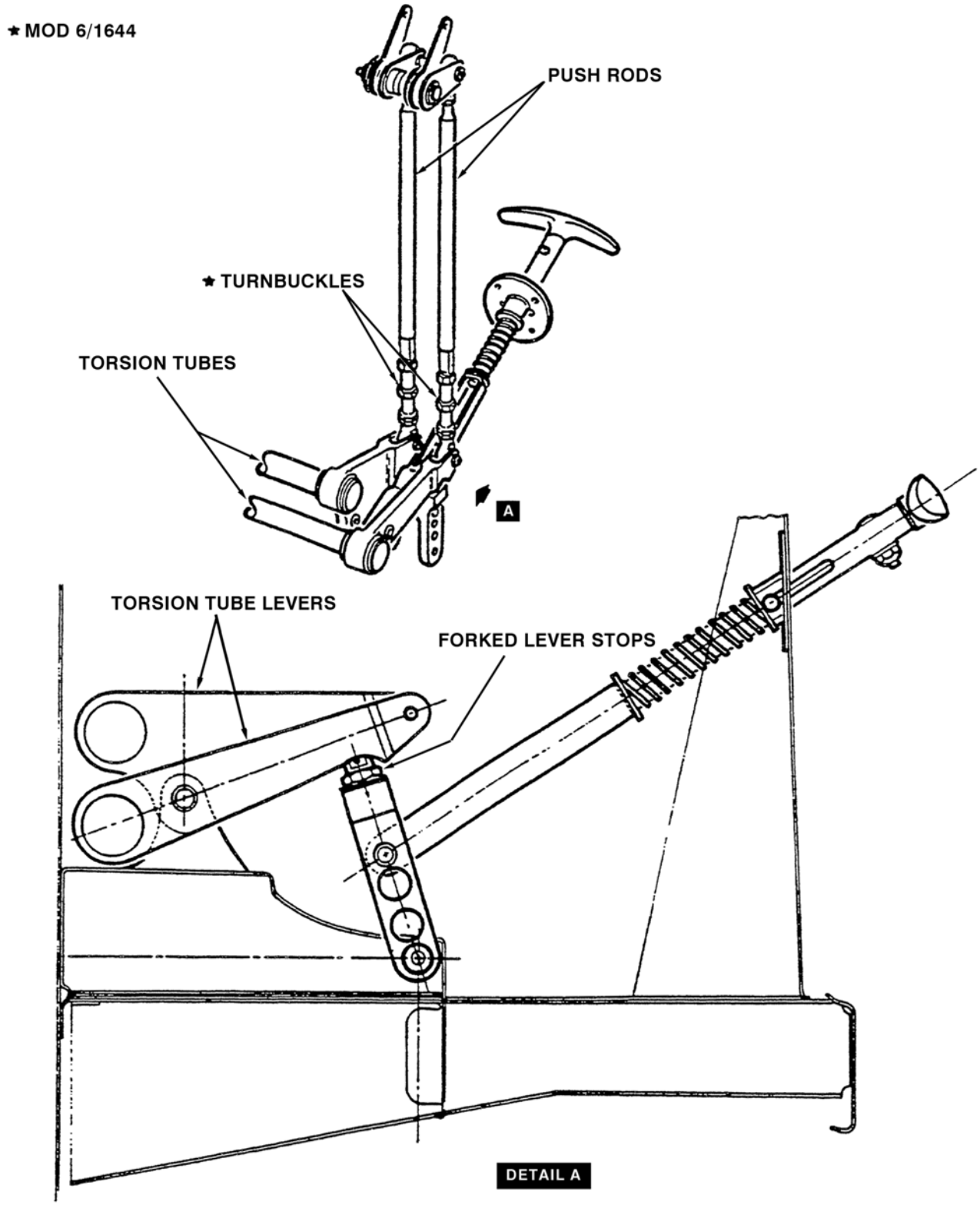
7) With system accumulator and wheel brakes accumulator fully charged, disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

8) Apply both brakes fully, pull up on parking brake handle and release brake pedals. Adjust fork lever stops to obtain a pressure reading of 850 ± 50 psi (refer to Figure 1).

9) Allow system to stand for 30 minutes. Check that system pressure is maintained. Release parking brake.

10) Remove pressure gauge from left and right-hand brake unit, and install bleeder plugs.

11) Set parking brake, if required.



Parking Brake
Figure 1

This Page Intentionally Left Blank